Type MS · Embedded systems: small computers inside devices

Answer 1 of 7 · Question 5 part a.ii · 2 marks

2025 JUN · Paper 01 · extract 1 of 5 · find part a.ii in this mark scheme extract

5(a)(i)	Award **one** mark for any of the following up to a maximum of **two** marks: • Heart rate (1) to measure blood flow/pulse rate (1) • Barometric pressure (1) to determine altitude (1) • Microphone (1) for voice activation control (1) • Temperature/Thermometer (1) to measure ambient/body temperature (1) • GPS (1) to calculate longitude/latitude (1) • Light (1) to adjust screen brightness/charge watch battery (1) • Magnetometer (1) to act as a compass (1) • ECG (1) measure heart rhythm (1) • Blood pressure monitor (1) measure pressure in arteries / can report heart rate/pulse rate (1)	Allow any reasonable sensor that might be embedded within a smart watch. Purpose must be linked to the sensor Purpose must expand on name (e.g. heart rate sensor - monitor user's heart rate is not sufficient for second mark) Allow purpose mentioned in second column if nothing in first column	**2**
Question	Answer	Additional Guidance	Mark
5(a)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: • It has a low power requirement (1) • It can use a low processor speed (1) • It has built in/integrated storage (1) • It relies on flash memory (1) • It can be mass produced at a low cost per device (1) • It is small (1) • It can work in a large range of operating temperature ranges (1) • It is robust (1) • It has its operating software set by the manufacturer (1) • It is designed to do one particular task (1)		**2**

Type MS · Embedded systems: small computers inside devices

Answer 1 of 7 · Question 5 part a.ii · 2 marks

2025 JUN · Paper 01 · extract 2 of 5 · find part a.ii in this mark scheme extract

Question	Answer	Additional Guidance	Mark
5(b)(i)	**The only correct answer is B** *A is not correct because low-level languages are not converted into high-level languages* *C is not correct because assemblers translate assembly code to produce machine code but do not execute it* *D is not correct because compilers and interpreters translate high-level languages*		1
Question	Answer	Additional Guidance	Mark
5(b)(ii)	Award up to **two** marks for a linked explanation such as: • Makes best use of the hardware (1) because they can directly access (specific) hardware (features) (1) • It reduces manufacturing costs (1) because of a lower processor specification/expense (1) • Optimise their code for hardware (1) directly access/control registers (1)	Do not accept references to translation time	2
Question	Answer	Additional Guidance	Mark
5(b)(iii)	Award **one** mark for any of the following up to a maximum of **two** marks: • Abide by the law / avoid plagiarism / copyright infringements (1) • Adhere to a professional code of conduct (1) • Maintain/enhance their skills through further learning/keep up to date/attend (relevant) conferences (1) • Test code thoroughly/do not release code with known errors/vulnerabilities (1) • Respect privacy and security of data (1) • Preserve the environment (1) • Make software accessible/unbiased (1)		2

Type MS · Embedded systems: small computers inside devices

Answer 1 of 7 · Question 5 part a.ii · 2 marks

2025 JUN · Paper 01 · extract 3 of 5 · find part a.ii in this mark scheme extract

Question	Answer	Additional Guidance	Mark
5(b)(iv)	**The only correct answer is A** *B is not correct because 500 KiB is 512,000 bytes* *C is not correct because 0.5 MiB is 524,288 bytes* *D is not correct because 5MB is 5,000,000 bytes*		**1**
Question	**Answer**	**Additional Guidance**	**Mark**
5(c)	Indicative content: • Can use sensors to take/track readings such as heart rate • Can track location via GPS and record routes taken/distance travelled • Can calculate exercise/movement undertaken such as steps/calories burned • Can be linked via 5G/Internet Of Things/Personal Area Network (PAN) to other devices/apps that can record and analyse the data • Can trigger alerts when users are inactive • Can monitor blood oxygen/blood sugar/sleep patterns • Can applaud/give kudos when an achievement/event takes place • Can increase motivation of the wearer to be healthy/perform exercise due to feedback loop • Workouts/alerts can be synced to the watch • Linked apps can perform more in-depth analysis • No guarantee that users will buy into benefits and lead a healthier lifestyle • Monitoring vulnerable people allowing independent living	Allow content referring to other wearable technology which is not a smartwatch	**6**
		Total for question 5	**16**

Type MS · Embedded systems: small computers inside devices

Answer 1 of 7 · Question 5 part a.ii · 2 marks

2025 JUN · Paper 01 · extract 4 of 5 · find part a.ii in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1-2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

Type MS · Embedded systems: small computers inside devices

Answer 1 of 7 · Question 5 part a.ii · 2 marks

2025 JUN · Paper 01 · extract 5 of 5 · find part a.ii in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Embedded systems: small computers inside devices

Answer 2 of 7 · Question 6 part a.i · 2 marks

2019 JUN · Paper 01 · extract 1 of 5 · find part a.i in this mark scheme extract

6(a)(i)	Features such as power consumption / processor / memory / number and type of components / physical size (1) can be optimised / chosen to fit just one specific task. (1)		2

Question Number	Answer	Additional Guidance	Mark
6(a)(ii)	Any **two** from: • Printer (1) • Bank card (magnetic strip) reader/scanner (1) • Bank card contactless/NFC reader/scanner (1) • Cash input and giving change device (1)	Allow any hardware that relates to paying by cash for MP4	2

Question Number	Answer	Additional Guidance	Mark
6(a)(iii)	To prevent unauthorised people from reading / understanding the bank card information (1)	Do not award 'to keep it from being hacked / accessed'.	1

Type MS · Embedded systems: small computers inside devices

Answer 2 of 7 · Question 6 part a.i · 2 marks

2019 JUN · Paper 01 · extract 2 of 5 · find part a.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)	Award **one** mark for each of (maximum of 4): 1. Readability: high-level language is easily readable by a human because it is designed to look like English whereas a low-level language is binary patterns / mnemonics / abbreviations. (1) 2. Portability: code written in a high-level language is easily transportable across machines whereas a low-level language is designed for a specific microprocessor / piece of hardware. (1) 3. Uses: low-level languages are commonly used to write programs for hardware or devices, like printers, whereas high-level languages are commonly used for to write higher-level applications, such as word processors. (1) 4. Optimisation: low-level languages are used to improve performance, especially of hardware, whereas high-level languages have to go through multiple layers of software to get to the hardware, e.g. the need for HLL to be compiled/interpreted makes programmes run slower than LLL, HLL has a higher level of abstraction removing unnecessary detail (one to many), LLL keeps more of a detail (one to one) (1) 5. Ease of use: high-level languages usually have tools, functions, libraries and development environments which are accessible, even by those with little knowledge, whereas, low-level languages are often challenging to work with and require expertise and depth of knowledge. (1)	For each mark point: • the general feature is followed by exemplification • needs to be a comparison, which may be implicit	4

Type MS · Embedded systems: small computers inside devices

Answer 2 of 7 · Question 6 part a.i · 2 marks

2019 JUN · Paper 01 · extract 3 of 5 · find part a.i in this mark scheme extract

Question Number	Answer	Mark
6(c)	Indicative content: Characteristics – Any of the following: • Implemented in software. • Could involve the use of neural networks / clustering / modelling • Algorithms are trained by exposure to data that has been validated / checked • Algorithms learn by identifying commonalities between the validated / checked data • Algorithms make predictions for new data based on prior learning • Algorithms may incorporate recursive self-improvement Uses – Any of the following: • Game playing (opponents) • Analytics (analyse buying patterns, predicting behaviours, predictive text, targeted advertising, personalised news feeds, friend suggestions) • Image / Audio processing (recognising objects/patterns, face recognition) • Logistics (scheduling, order fulfilment) • Control systems (cars, manufacturing, weapons, navigation • Diagnostic systems (medical, mechanical, electrical) • Natural languages processing (chatterbots, chatbots, speech recognition, personal assistants) • Robotics (dangerous situations, help aged or disabled) Ethical issues – Any of the following: • Training sets could include data that should not be used to make decisions (race, religion, gender, etc.) that could cause biased outcomes • Is a computer to be trusted to make decisions (life-death)? • If a computer discovers something that humans can't prove, should it be accepted as truth? • How do we know that the answer produced is correct? • Will people be comfortable interacting with machines that are considered intelligent? • Some jobs may be at risk, especially those requiring human skills such as empathy or creativity.	6

Type MS · Embedded systems: small computers inside devices

Answer 2 of 7 · Question 6 part a.i · 2 marks

2019 JUN · Paper 01 · extract 4 of 5 · find part a.i in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent and logically structured.

Type MS · Embedded systems: small computers inside devices

Answer 2 of 7 · Question 6 part a.i · 2 marks

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Type MS · Embedded systems: small computers inside devices

Answer 3 of 7 · Question 6 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 1 of 5 · find part a.ii in this mark scheme extract

6(a)(i)	Features such as power consumption / processor / memory / number and type of components / physical size (1) can be optimised / chosen to fit just one specific task. (1)		2

Question Number	Answer	Additional Guidance	Mark
6(a)(ii)	Any **two** from: • Printer (1) • Bank card (magnetic strip) reader/scanner (1) • Bank card contactless/NFC reader/scanner (1) • Cash input and giving change device (1)	Allow any hardware that relates to paying by cash for MP4	2

Question Number	Answer	Additional Guidance	Mark
6(a)(iii)	To prevent unauthorised people from reading / understanding the bank card information (1)	Do not award 'to keep it from being hacked / accessed'.	1

Type MS · Embedded systems: small computers inside devices

Answer 3 of 7 · Question 6 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 2 of 5 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)	Award **one** mark for each of (maximum of 4): 1. Readability: high-level language is easily readable by a human because it is designed to look like English whereas a low-level language is binary patterns / mnemonics / abbreviations. (1) 2. Portability: code written in a high-level language is easily transportable across machines whereas a low-level language is designed for a specific microprocessor / piece of hardware. (1) 3. Uses: low-level languages are commonly used to write programs for hardware or devices, like printers, whereas high-level languages are commonly used for to write higher-level applications, such as word processors. (1) 4. Optimisation: low-level languages are used to improve performance, especially of hardware, whereas high-level languages have to go through multiple layers of software to get to the hardware, e.g. the need for HLL to be compiled/interpreted makes programmes run slower than LLL, HLL has a higher level of abstraction removing unnecessary detail (one to many), LLL keeps more of a detail (one to one) (1) 5. Ease of use: high-level languages usually have tools, functions, libraries and development environments which are accessible, even by those with little knowledge, whereas, low-level languages are often challenging to work with and require expertise and depth of knowledge. (1)	For each mark point: • the general feature is followed by exemplification • needs to be a comparison, which may be implicit	4

Type MS · Embedded systems: small computers inside devices

Answer 3 of 7 · Question 6 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 3 of 5 · find part a.ii in this mark scheme extract

Question Number	Answer	Mark
6(c)	Indicative content: Characteristics – Any of the following: • Implemented in software. • Could involve the use of neural networks / clustering / modelling • Algorithms are trained by exposure to data that has been validated / checked • Algorithms learn by identifying commonalities between the validated / checked data • Algorithms make predictions for new data based on prior learning • Algorithms may incorporate recursive self-improvement Uses – Any of the following: • Game playing (opponents) • Analytics (analyse buying patterns, predicting behaviours, predictive text, targeted advertising, personalised news feeds, friend suggestions) • Image / Audio processing (recognising objects/patterns, face recognition) • Logistics (scheduling, order fulfilment) • Control systems (cars, manufacturing, weapons, navigation • Diagnostic systems (medical, mechanical, electrical) • Natural languages processing (chatterbots, chatbots, speech recognition, personal assistants) • Robotics (dangerous situations, help aged or disabled) Ethical issues – Any of the following: • Training sets could include data that should not be used to make decisions (race, religion, gender, etc.) that could cause biased outcomes • Is a computer to be trusted to make decisions (life-death)? • If a computer discovers something that humans can't prove, should it be accepted as truth? • How do we know that the answer produced is correct? • Will people be comfortable interacting with machines that are considered intelligent? • Some jobs may be at risk, especially those requiring human skills such as empathy or creativity.	6

Type MS · Embedded systems: small computers inside devices

Answer 3 of 7 · Question 6 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 4 of 5 · find part a.ii in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent and logically structured.

Type MS · Embedded systems: small computers inside devices

Answer 3 of 7 · Question 6 part a.ii · 2 marks

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Type MS · Embedded systems: small computers inside devices

Answer 4 of 7 · Question 5 part c.i · 2 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part c.i in this mark scheme extract

5(a)	Award **one** mark for each correct component: **Component** / **Letter** Control unit — B Output device — G Registers — C Main memory — H Cache — F Input device — E Arithmetic/logic unit (ALU) — D Clock — A		5

Type MS · Embedded systems: small computers inside devices

Answer 4 of 7 · Question 5 part c.i · 2 marks

2021 NOV · Paper 01 · extract 2 of 4 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(i)	Award **one** mark each to a maximum of **two** marks for: • stores a memory address (for the next instruction) (1) • memory address incremented/changed during the fetch process (1)		2
5(b)(ii)	**The only correct answer is B** A *is not correct because* there is no such signal C *is not correct because* as there is no such signal D *is not correct because* as that does not take place during the fetch stage		1
5(b)(iii)	Award **one** mark from: • processes more instructions in the same amount of time (1) • more cycles can be performed in the same amount of time (1) • fetch-execute-decode cycle runs faster (1)		1

Type MS · Embedded systems: small computers inside devices

Answer 4 of 7 · Question 5 part c.i · 2 marks

2021 NOV · Paper 01 · extract 3 of 4 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(c)(i)	Award **one** mark from: • a system designed for one/a few specific functions/task(s) (1) • has both hardware and software (1) • has integrated memory (1) • has no or minimal user interface (1) • is power efficient/low power consumption (1) • its functionality cannot be changed/upgraded by users (1) • often use sensors and actuators to interact with the external environment (1) • functions in real time (1)		2
5(c)(ii)	Award up to **two** marks for a linked explanation such as: • Performance / number of cores / power consumption needs to be adequate for the task (1) because any more would waste money / power (1) • Size of cache can be small or non-existent (1) because there is no need to store frequently used instructions (1) • The RAM can be limited/very small (1) because it is only holding a limited number of instructions (1) • Only needs a small bus width (1) because the output signals are serial binary (1)		2
		Total	**13**

Type MS · Embedded systems: small computers inside devices

Answer 4 of 7 · Question 5 part c.i · 2 marks

2021 NOV · Paper 01 · extract 4 of 4 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Embedded systems: small computers inside devices

Answer 5 of 7 · Question 5 part c.ii · 2 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part c.ii in this mark scheme extract

5(a)	Award **one** mark for each correct component: **Component** / **Letter** Control unit — B Output device — G Registers — C Main memory — H Cache — F Input device — E Arithmetic/logic unit (ALU) — D Clock — A		5

Type MS · Embedded systems: small computers inside devices

Answer 5 of 7 · Question 5 part c.ii · 2 marks

2021 NOV · Paper 01 · extract 2 of 4 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(i)	Award **one** mark each to a maximum of **two** marks for: • stores a memory address (for the next instruction) (1) • memory address incremented/changed during the fetch process (1)		2
5(b)(ii)	**The only correct answer is B** A *is not correct because* there is no such signal C *is not correct because* as there is no such signal D *is not correct because* as that does not take place during the fetch stage		1
5(b)(iii)	Award **one** mark from: • processes more instructions in the same amount of time (1) • more cycles can be performed in the same amount of time (1) • fetch-execute-decode cycle runs faster (1)		1

Type MS · Embedded systems: small computers inside devices

Answer 5 of 7 · Question 5 part c.ii · 2 marks

2021 NOV · Paper 01 · extract 3 of 4 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(c)(i)	Award **one** mark from: • a system designed for one/a few specific functions/task(s) (1) • has both hardware and software (1) • has integrated memory (1) • has no or minimal user interface (1) • is power efficient/low power consumption (1) • its functionality cannot be changed/upgraded by users (1) • often use sensors and actuators to interact with the external environment (1) • functions in real time (1)		2
5(c)(ii)	Award up to **two** marks for a linked explanation such as: • Performance / number of cores / power consumption needs to be adequate for the task (1) because any more would waste money / power (1) • Size of cache can be small or non-existent (1) because there is no need to store frequently used instructions (1) • The RAM can be limited/very small (1) because it is only holding a limited number of instructions (1) • Only needs a small bus width (1) because the output signals are serial binary (1)		2
		Total	**13**

Type MS · Embedded systems: small computers inside devices

Answer 5 of 7 · Question 5 part c.ii · 2 marks

2021 NOV · Paper 01 · extract 4 of 4 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Embedded systems: small computers inside devices

Answer 6 of 7 · Question 5 part c · 2 marks

2023 JUN · Paper 01 · extract 1 of 8 · find part c in this mark scheme extract

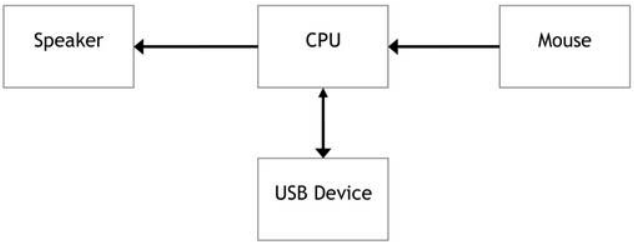
<table>
<tr>
<td>5(a)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Req.</th><th>Example</th></tr>
<tr><td>Input</td><td>1. Enter the direction the bee will face (1)
2. Enter the number of steps that the bee will move (1)</td></tr>
<tr><td>Process</td><td>Calculate the path the bee will move along to its new position</td></tr>
<tr><td>Output</td><td>3. The bee appears in its new position (1)
4. The bee flies to its new position (1)
5. The bee faces its new direction (1)</td></tr>
</table>
</td>
<td>Allow direct instructions as input e.g. ‘Turn East’ and consequences for output e.g. ‘Bee turns to face East’ if clear.

The input/output do not have to be directly linked. E.g. Input Turn West, Output Has Moved 20 steps</td>
<td>2</td>
</tr>
</table>

Type MS · Embedded systems: small computers inside devices

Answer 6 of 7 · Question 5 part c · 2 marks

2023 JUN · Paper 01 · extract 2 of 8 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5b(i)	Award one mark for each: • Line between Mouse and CPU with arrow at CPU end only (1) • Line between CPU and Speaker with arrow at Speaker end only (1) • Line between CPU and USB Device with arrows on both ends (1) 	Allow two independent back/forth arrows between CPU and USB in place of a single bi-directional arrow. Ignore any additional lines.	3

Type MS · Embedded systems: small computers inside devices

Answer 6 of 7 · Question 5 part c · 2 marks

2023 JUN · Paper 01 · extract 3 of 8 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(ii)	**The only correct answer is B** ***A*** *is not correct because a* ***firewall*** *is an application that is loaded after the device is booted* ***C*** *is not correct because* ***the operating system*** *is loaded after the device is booted* ***D*** *is not correct because* ***the user interface*** *is loaded after the device is booted*		1

Question Number	Answer	Additional Guidance	Mark
5(b)(iii)	Award up to **two** marks for a linked explanation, such as: • Cache makes up for the difference in speed of two devices (1) because it is very high-speed memory (1) • Cache is high-speed memory (1) which acts as a buffer between a faster and a slower device (1) • It speeds up processing (1) because active/commonly used instructions and data are stored nearer to the CPU / the CPU doesn't have to wait while instructions/data are fetched from RAM (1)		2

Type MS · Embedded systems: small computers inside devices

Answer 6 of 7 · Question 5 part c · 2 marks

2023 JUN · Paper 01 · extract 4 of 8 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(c)	Award up to **two** marks for a linked description, such as: • Uses a sensor (1) to check the water level in the machine (1) • Uses a thermometer (1) to check the water temperature (1) • Uses a timer (1) to control the time required for the wash cycle/rinse/drain/spin (1) • Uses a motor (1) to rotate the drum (1)	The first mark is for identification and the second is for the linked description. It is possible to gain one mark for a description without identifying the sensor/actuator.	2

Type MS · Embedded systems: small computers inside devices

Answer 6 of 7 · Question 5 part c · 2 marks

2023 JUN · Paper 01 · extract 5 of 8 · find part c in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(i)</td><td>Award one mark for each two correct rows.

<table>
<tr><th>A</th><th>B</th><th>C</th><th>A AND C</th><th>(A AND C) OR B</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr>
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<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr>
</table>
</td><td></td><td>3</td></tr>
</table>

Type MS · Embedded systems: small computers inside devices

Answer 6 of 7 · Question 5 part c · 2 marks

2023 JUN · Paper 01 · extract 6 of 8 · find part c in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(ii)</td><td>Award one mark for each of:

• M OR P (1)
• NOT C (1)
• <expr1> AND <expr2> (1)

Example statements:

(M OR P) AND (NOT C)
(M OR P) AND NOT C
(M AND NOT C) OR (P AND NOT C)
M AND NOT C OR P AND NOT C

For information only:

<table>
<tr><th>M</th><th>P</th><th>C</th><th>M OR P</th><th>NOT C</th><th>(M OR P) AND (NOT C)</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
</table>
</td><td>Third bullet awarded even if first two are not

Order of precedence is NOT, AND, OR

Check brackets carefully, if provided</td><td>3</td></tr>
</table>

Type MS · Embedded systems: small computers inside devices

Answer 6 of 7 · Question 5 part c · 2 marks

2023 JUN · Paper 01 · extract 7 of 8 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(e)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. The operating system uses a scheduler to control processes (1) 2. The operating system holds processes in a queue (1) 3. Some processes may be given higher priorities than others (1) 4. Each process gains accesses to the CPU for a short time / time slice to execute (1) 5. Processes are swapped to/from (the queue / CPU)	Do not award responses about memory management, as the question is about sharing the CPU only	2

Type MS · Embedded systems: small computers inside devices

Answer 6 of 7 · Question 5 part c · 2 marks

2023 JUN · Paper 01 · extract 8 of 8 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Embedded systems: small computers inside devices

Answer 7 of 7 · Question 5 part a.i · 2 marks

2025 JUN · Paper 01 · extract 1 of 5 · find part a.i in this mark scheme extract

5(a)(i)	Award **one** mark for any of the following up to a maximum of **two** marks: • Heart rate (1) to measure blood flow/pulse rate (1) • Barometric pressure (1) to determine altitude (1) • Microphone (1) for voice activation control (1) • Temperature/Thermometer (1) to measure ambient/body temperature (1) • GPS (1) to calculate longitude/latitude (1) • Light (1) to adjust screen brightness/charge watch battery (1) • Magnetometer (1) to act as a compass (1) • ECG (1) measure heart rhythm (1) • Blood pressure monitor (1) measure pressure in arteries / can report heart rate/pulse rate (1)	Allow any reasonable sensor that might be embedded within a smart watch. Purpose must be linked to the sensor Purpose must expand on name (e.g. heart rate sensor - monitor user's heart rate is not sufficient for second mark) Allow purpose mentioned in second column if nothing in first column	**2**
Question	Answer	Additional Guidance	Mark
5(a)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: • It has a low power requirement (1) • It can use a low processor speed (1) • It has built in/integrated storage (1) • It relies on flash memory (1) • It can be mass produced at a low cost per device (1) • It is small (1) • It can work in a large range of operating temperature ranges (1) • It is robust (1) • It has its operating software set by the manufacturer (1) • It is designed to do one particular task (1)		**2**

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Answer 7 of 7 · Question 5 part a.i · 2 marks

2025 JUN · Paper 01 · extract 2 of 5 · find part a.i in this mark scheme extract

Question	Answer	Additional Guidance	Mark
5(b)(i)	**The only correct answer is B** *A is not correct because low-level languages are not converted into high-level languages* *C is not correct because assemblers translate assembly code to produce machine code but do not execute it* *D is not correct because compilers and interpreters translate high-level languages*		1
Question	Answer	Additional Guidance	Mark
5(b)(ii)	Award up to **two** marks for a linked explanation such as: • Makes best use of the hardware (1) because they can directly access (specific) hardware (features) (1) • It reduces manufacturing costs (1) because of a lower processor specification/expense (1) • Optimise their code for hardware (1) directly access/control registers (1)	Do not accept references to translation time	2
Question	Answer	Additional Guidance	Mark
5(b)(iii)	Award **one** mark for any of the following up to a maximum of **two** marks: • Abide by the law / avoid plagiarism / copyright infringements (1) • Adhere to a professional code of conduct (1) • Maintain/enhance their skills through further learning/keep up to date/attend (relevant) conferences (1) • Test code thoroughly/do not release code with known errors/vulnerabilities (1) • Respect privacy and security of data (1) • Preserve the environment (1) • Make software accessible/unbiased (1)		2

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Answer 7 of 7 · Question 5 part a.i · 2 marks

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Question	Answer	Additional Guidance	Mark
5(b)(iv)	**The only correct answer is A** *B is not correct because 500 KiB is 512,000 bytes* *C is not correct because 0.5 MiB is 524,288 bytes* *D is not correct because 5MB is 5,000,000 bytes*		**1**
Question	**Answer**	**Additional Guidance**	**Mark**
5(c)	Indicative content: • Can use sensors to take/track readings such as heart rate • Can track location via GPS and record routes taken/distance travelled • Can calculate exercise/movement undertaken such as steps/calories burned • Can be linked via 5G/Internet Of Things/Personal Area Network (PAN) to other devices/apps that can record and analyse the data • Can trigger alerts when users are inactive • Can monitor blood oxygen/blood sugar/sleep patterns • Can applaud/give kudos when an achievement/event takes place • Can increase motivation of the wearer to be healthy/perform exercise due to feedback loop • Workouts/alerts can be synced to the watch • Linked apps can perform more in-depth analysis • No guarantee that users will buy into benefits and lead a healthier lifestyle • Monitoring vulnerable people allowing independent living	Allow content referring to other wearable technology which is not a smartwatch	**6**
		Total for question 5	**16**

Type MS · Embedded systems: small computers inside devices

Answer 7 of 7 · Question 5 part a.i · 2 marks

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Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1-2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

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Answer 7 of 7 · Question 5 part a.i · 2 marks

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Question	Answer	Additional Guidance	Mark